Lab 4: VibeCoding — Chat History

Mahipal S · PES1UG24AM153 · Repo #28 Defender (SETAPESU26/28_defender) · LLM used: Claude (claude.ai) · 22 Oct 2026

Prompt 1 (user): i am mahipal s, pes1ug24am153, i want u to do my se lab 4 [attached: repo allocation list, Lab 4 handout]

Claude: Matched SRN PES1UG24AM153 to repo #28 (Defender). Cloned the repo and read the README: 1 bug (radar) + 3 features (sky_color, on_humanoid_rescued, bonus_life_threshold). Fixed and implemented each task, one commit per task, and checked each one with a headless test.

Task 1 — Fix radar bug (commit 76c2e97)

draw_radar used self.screen_x(x) % VIEW_W, which is the player-relative screen position, so blips clustered around the player. Changed it to the absolute world position scaled to the radar width.

```
Task 1: Fix radar to plot blips at absolute world position

diff --git a/game.py b/game.py
index 4a96440..640bf49 100644
--- a/game.py
+++ b/game.py
@@ -203,7 +203,9 @@ class Game:
         blips += [(l.x, l.y, (255, 90, 90) if l.mutant else (230, 200, 60)) for l in self.landers]
         blips.append((self.player.x, self.player.y, (255, 255, 255)))
         for x, y, color in blips:
-            rx = self.screen_x(x) % VIEW_W
+            # Map absolute world x (0..WORLD_W) onto the full radar width (0..VIEW_W),
+            # independent of where the player is.
+            rx = (x % WORLD_W) / WORLD_W * VIEW_W
             ry = (y - PLAY_TOP) / (VIEW_H - PLAY_TOP) * (RADAR_H - 8) + 4
             pygame.draw.rect(screen, color, (rx - 2, ry - 2, 4, 4))
```

Task 2 — sky_color(wave) (commit f22ff85)

Wave 1 returns None (default sky). Later waves cycle through blue, purple, crimson, teal and amber, getting slightly brighter each full cycle.

```
Task 2: Implement sky_color to change sky hue each wave

diff --git a/game.py b/game.py
index 640bf49..f7232b5 100644
--- a/game.py
+++ b/game.py
@@ -13,7 +13,19 @@ PHASES = [random.uniform(0, math.tau) for _ in range(3)]

 def sky_color(wave):
     """Return an (r, g, b) sky colour for the current wave, or None for the default."""
-    pass
+    if wave <= 1:
+        return None  # keep the default near-black sky for the first wave
+    # Each new wave shifts the hue and gets slightly brighter (capped so it stays dark).
+    palette = [
+        (10, 10, 45),   # deep blue
+        (30, 10, 50),   # purple
+        (45, 12, 30),   # crimson
+        (10, 35, 40),   # teal
+        (40, 28, 10),   # amber
+    ]
+    r, g, b = palette[(wave - 2) % len(palette)]
+    boost = min(30, (wave - 2) // len(palette) * 10)
+    return (min(255, r + boost), min(255, g + boost), min(255, b + boost))


 def on_humanoid_rescued(humanoid):
```

Task 3 — on_humanoid_rescued(humanoid) (commit 94a3b12)

Adds a floating '+500' popup at the rescue spot. A module-level POPUPS list is aged in update(), drawn in draw(), and cleared on reset.

```
Task 3: Implement on_humanoid_rescued with floating +500 popup

diff --git a/game.py b/game.py
index f7232b5..061b3b4 100644
--- a/game.py
+++ b/game.py
@@ -9,6 +9,7 @@ PLAY_TOP = RADAR_H + 6
 HUMANOID_COUNT = 6
```

```
 FALL_LIMIT = 160
 PHASES = [random.uniform(0, math.tau) for _ in range(3)]
+POPUPS = []  # floating score popups: {"x", "y", "life"}


 def sky_color(wave):
@@ -30,7 +31,8 @@ def sky_color(wave):

 def on_humanoid_rescued(humanoid):
     """Called when the player catches a falling humanoid; add a bonus or celebration here."""
-    pass
+    # Spawn a floating "+500" popup at the rescue spot (world coordinates).
+    POPUPS.append({"x": humanoid.x, "y": humanoid.y - 20, "life": 1.2})


 def bonus_life_threshold():
@@ -139,6 +141,7 @@ class Game:
         self.landers, self.bullets = [], []
         self.score, self.lives, self.wave, self.state = 0, 3, 1, "play"
         self.bonus_awarded = 0
+        POPUPS.clear()
         self.start_wave()

     def start_wave(self):
@@ -182,6 +185,11 @@ class Game:
                 self.score += 500
                 on_humanoid_rescued(humanoid)
         self.update_bullets(dt)
+        for popup in POPUPS[:]:
+            popup["y"] -= 40 * dt
+            popup["life"] -= dt
+            if popup["life"] <= 0:
+                POPUPS.remove(popup)
         if player.invulnerable <= 0:
             for lander in self.landers:
                 if abs(wrap_delta(player.x, lander.x)) < 22 and abs(lander.y - player.y) < 18:
@@ -242,6 +250,11 @@ class Game:
         if player.invulnerable <= 0 or int(player.invulnerable * 10) % 2 == 0:
             f, cx = player.facing, VIEW_W / 2
             pygame.draw.polygon(screen, (240, 240, 250), [(cx + f * 18, player.y), (cx - f * 14, player.y - 8), (cx - f *
+        for popup in POPUPS:
+            sx = self.screen_x(popup["x"])
+            if -40 < sx < VIEW_W + 40:
+                label = self.font.render("+500", True, (255, 230, 90))
+                screen.blit(label, label.get_rect(center=(sx, popup["y"])))
         self.draw_radar(screen)
         hud = self.font.render(f"Score {self.score}  Lives {self.lives}  Wave {self.wave}  Humanoids {len(self.humanoids)}
         screen.blit(hud, (10, RADAR_H + 4))
```

Task 4 — bonus_life_threshold() (commit c79e31b)

Returns 10000, so the existing bookkeeping awards one extra life per 10,000 points.

```
Task 4: Implement bonus_life_threshold (extra life every 10000 points)

diff --git a/game.py b/game.py
index 061b3b4..e375ddf 100644
--- a/game.py
+++ b/game.py
@@ -37,7 +37,7 @@ def on_humanoid_rescued(humanoid):

 def bonus_life_threshold():
     """Return a score value at which the player earns an extra life, or None to disable bonus lives."""
-    pass
+    return 10000  # one extra life every 10,000 points


 def wrap_delta(a, b):
```

Verification output: radar blips for world x=0/1600/3199 sit at 0/400/800 px wherever the player is; rescue gives +500 and a popup that expires; lives 3→4 at 10,000 with no double award.

Prompt 2 (user): no, i want u to retry and do everything for me, even record, pls do it!

Claude: Recorded 10-second before/after gameplay clips by running game.py headlessly with scripted input. The original commit (031f1ee) is used for 'before' and the final commit for 'after'. In the after clip, the score starts at 9,400 and one lander carrying a humanoid is placed in front of the player, so the rescue popup, wave-2 sky change and bonus life all appear within 10 seconds.